Process Characterization Report

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCR-003

Process Development / Manufacturing Science & Technology (MSAT)

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Abbreviations

ADCC, antibody dependent cellular cytotoxicity. AMV, analytical method validation. ANOVA, analysis of variance. ATP, adenosine triphosphate. BLA, Biologics License Application. CCD, central composite design. CHO, Chinese hamster ovary. CPP, critical process parameter. CPV, continued process verification. CQA, critical quality attribute. DoE, design of experiments. ELISA, enzyme linked immunosorbent assay. GDP, guanosine diphosphate. GlcNAc, N-acetylglucosamine. GPP, general process parameter. HCP, host cell protein. HILIC, hydrophilic interaction liquid chromatography. HMW, high molecular weight. icIEF, imaged capillary isoelectric focusing. KPP, key process parameter. LOQ, limit of quantitation. MSAT, Manufacturing Science and Technology. NOR, normal operating range. PAR, proven acceptable range. pCO2, dissolved carbon dioxide partial pressure. qPCR, quantitative polymerase chain reaction. RSD, relative standard deviation. RSM, response surface methodology. SDM, scale-down model. SEC, size exclusion chromatography. SOP, standard operating procedure. UDP, uridine diphosphate. UPLC, ultra performance liquid chromatography. VCD, viable cell density. WC-CPP, well controlled critical process parameter.

Executive summary

The production bioreactor is the fed-batch cell culture step of the A-Mab drug substance process and it operates at a working volume of 15,000 L. This report presents the characterization study executed on the qualified scale-down model under the plan PCP-003. The study quantified the relationship between the culture parameters and the quality attributes formed at this step. From that relationship it defines the operating region, the proven acceptable ranges, the commercial-scale capability and the parameter classification that the step contributes to the control strategy.

The culture forms 7 of the drug substance quality attributes (Table 2.1). Five of them were measured as responses in the designed experiments. The other two, host cell protein and residual DNA, enter the broth when cells lyse, and both rise as viability falls through the last days of the run. Of the 9 process parameters carried into the study, 5 were studied in a multivariate design and 4 were assessed one at a time. The multivariate work ran in two stages. A 19-run screening design identified the active factors and their interactions, and a 28-run face-centred central composite design in the four factors carrying the largest effects produced the predictive models used for the design space and the proven acceptable ranges.

Culture duration and culture pH carry the largest effects on the glycan attributes, and dissolved carbon dioxide carries the largest effect on acidic charge variants. The interaction between culture pH and culture duration is significant for both afucosylation and galactosylation, so neither parameter can be set without reference to the other. The response-surface models account for between 91 and 97 % of the variance in the five responses. No response shows significant lack of fit against the centre-point pure error.

The design space is the region of the four modelled parameters over which every attribute governed here meets its in-process criterion. It covers 57 % of the characterized region, and galactosylation is the attribute that rejects most of the remainder. Commercial-scale capability was estimated from 2,000 simulated batches with every parameter varying inside its normal operating range. All 7 attributes meet their drug substance acceptance criteria, and the tightest capability is high mannose at a one-sided Cpk of 1.94.

Two results bound the operating claim. At the corners of the normal operating range that combine the low end of the pH range with the shortest culture, the model predicts afucosylation of 9.431 % against an in-process alert limit of 9.426 %, and galactosylation of 31.8 % against an alert limit of 30.1 %. Both corners stay inside the drug substance acceptance criteria by a wide margin. The alert limits are set from demonstrated capability and are explained in §7.

Of the 9 parameters, 5 were classified as well controlled critical process parameters, 3 as key process parameters and 1 as a general process parameter. No parameter at this step was classified as a critical process parameter. Both of the 2 deviations recorded during execution were retained (§13). Neither changed a conclusion of the study. The outcome of this report is rolled up into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose primary mechanism of action is antibody dependent cellular cytotoxicity. It is expressed in a recombinant CHO cell line in fed-batch culture and purified on a platform train of Protein A capture, low pH viral inactivation, cation exchange chromatography, anion exchange chromatography, small-virus retentive filtration and ultrafiltration with diafiltration (CMC Biotech Working Group 2009). The production bioreactor is Step 3 of that process and is the subject of this report.

The production bioreactor is the only step of the drug substance process at which product quality attributes are formed. The glycan distribution is completed in the Golgi apparatus before the antibody is secreted, and the charge variant distribution is established inside the cell and in the culture fluid. The platform purification train does not modify them.

The culture is inoculated from the N-1 seed bioreactor, fed with a single nutrient feed during the run and harvested after about 17 days. The pH, the temperature, the dissolved gases and the osmolality of the medium set the rate at which the cells grow, the antibody each cell secretes and the post-translational modifications that antibody carries. A cooler culture grows more slowly and stays productive for longer. A warmer culture deamidates its product faster in the broth, and a more alkaline one aggregates it faster. The step forms the three glycan attributes, the charge variant distribution and the aggregate level of the harvested pool. It also loads the harvest with host cell protein and DNA, both of which are released from lysed cells as viability falls late in the run and are cleared by the purification train.

1.2 Objectives

The study had five objectives.

(i) To quantify the relationship between the production bioreactor process parameters and the quality attributes formed at this step, over ranges wider than the routine control ranges.
(ii) To define the multivariate operating region within which those attributes meet the criteria that apply to this step.
(iii) To establish a proven acceptable range for each parameter that governs a quality attribute, both at the set-point and with normal operating range variation propagated.

(iv) To classify every process parameter of the step from its demonstrated effect and from the capability with which it is controlled.
(v) To justify the Stage 2 operating ranges and state this step's contribution to the control strategy of the drug substance process.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced development approach of ICH Q8 and ICH Q11, in which prior knowledge and a risk assessment direct the experimental work and the result is a design space and a control strategy (International Council for Harmonisation 2009) (International Council for Harmonisation 2012). Criticality is treated as a continuum rather than as a binary classification, in line with ICH Q9 (International Council for Harmonisation 2023b). The process model, the quality attribute framework and the risk methodology follow the A-Mab case study (CMC Biotech Working Group 2009). No viral clearance claim is made for this step, so ICH Q5A does not apply to it (International Council for Harmonisation 2023a).

This report is one of the per-step reports of the characterization campaign. Its scope was set by the pre-characterization risk assessment RA-001, its execution was prescribed by the plan PCP-003, and its results are consolidated in PCMR-001. Table 1.1 lists the documents this report is bound to.

Table 1.1: Documents related to this report.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-003	Process Characterization Plan (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The A-Mab production bioreactor uses the Novacyte humanized IgG1 platform, which has produced three previous licensed antibodies (X-Mab, Y-Mab and Z-Mab) in the same fed-batch format. The platform contributes the medium and feed system, the control strategy for pH, temperature, dissolved oxygen and dissolved carbon dioxide, and a long performance history at commercial scale (CMC Biotech Working Group 2009). Two of the A-Mab acceptance criteria are set from that history rather than from A-Mab data alone. The upper limit for aggregate of 5 % and the upper limit for host cell protein of 100 ng/mg both come from X-Mab clinical experience.

The mechanisms by which the culture parameters act on the glycan attributes are established for the platform. Core fucose and terminal galactose are added to the biantennary glycan in the Golgi by a fucosyltransferase and a galactosyltransferase, each working from a nucleotide sugar donor. The activity of those enzymes, the supply of the donors and the residence time of the glycan in the Golgi therefore set the fraction of antibody that leaves the cell without core fucose and the fraction that leaves galactosylated. A rise in culture pH raises intracellular pH and makes the Golgi lumen less acidic, and the transferases work below neutral pH, so less galactose is transferred and more glycan leaves the cell unprocessed. A rise in culture temperature runs every enzymatic step faster, so more glycan is galactosylated before the antibody is secreted. Dissolved carbon dioxide crosses the plasma membrane and hydrates to carbonic acid, which lowers cytosolic pH, and the bicarbonate that neutralizes it accumulates in the medium and raises osmolality. High mannose glycans are the species on which the mannosidases and the later transferases never acted, so their fraction rises whenever the glycan leaves the Golgi before those enzymes have finished.

Culture duration acts on the harvested pool rather than on the cell. The pool is the sum of everything secreted over the run, so a longer culture weights it toward antibody secreted after the UDP-galactose donor and the manganese cofactor of the galactosyltransferase have been drawn down. Galactosylation was therefore expected to fall as the culture ages. Culture pH sets how fast the galactosyltransferase works and culture duration sets how much of the pool that enzyme made while its donor was still plentiful, so an interaction between the two was expected in both glycan responses, and both parameters were carried into the response-surface design.

Aggregate and acidic charge variants are formed by chemistry in the culture fluid rather than in the Golgi. Aggregate arises from antibody that partially unfolds or associates while it sits at culture temperature for days, so its level rises with the time the product spends in the reactor. A culture pH nearer the isoelectric point of the antibody leaves less net charge on the molecule, the electrostatic repulsion between two molecules falls with that charge, and aggregate rises. Acidic charge variants are mainly the products of asparagine deamidation, whose rate rises with pH and with temperature, together with sialylated glycans and glycated lysines. A parameter that raises deamidation may lower the sialylated fraction at the same time, so the net direction of each parameter on the sum was read from the fitted model.

Host cell protein and residual DNA have a common origin. Both are released from the cytosol and the nucleus when cells lyse, and viability falls through the late phase of a fed-batch culture, so both rise as the culture ages and the composition of the load shifts from secreted proteins toward intracellular ones. The purification train reduces both by orders of magnitude, so neither is governed by an acceptance criterion at this step.

2.2 Quality attributes in scope

Table 2.1 lists the quality attributes this step sets, with the acceptance criterion, the criticality level and the Tool #1 score behind that level. Tool #1 scores the impact of an attribute on safety and efficacy against the uncertainty in that relationship, and the product of the two places each attribute on a continuum of criticality rather than in a binary class (CMC Biotech Working Group 2009). Attributes scored high or very high are treated as critical.

Table 2.1: Quality attributes set by the production bioreactor, with acceptance criteria and criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6

High mannose carries the highest Tool #1 score of the group at 80, because high mannose species clear faster from the circulation and the exposure relationship carries the larger uncertainty. Afucosylation and galactosylation are critical through effector function, which is the primary mechanism of action of A-Mab. Aggregate is critical through immunogenicity. Acidic charge variants and residual DNA carry the lowest Tool #1 scores of the group.

2.3 Risk-based prioritization and parameter selection

The scope of this study was set by RA-001, which ranked every parameter of the step on its potential impact on a quality attribute and on its potential to interact with other parameters, and then filtered the ranked list into study types (CMC Biotech Working Group 2009). Parameters that scored high on both axes were assigned to the multivariate design. Parameters that scored high on impact but low on interaction, or that are held by an independent control loop, were assigned to univariate assessment. Of the 9 parameters, 5 were assigned to the multivariate design and 4 to univariate assessment. No parameter of this step was excluded from study.

The ranges studied were widened beyond the routine control ranges so that the models would cover the settings a commercial batch can reach, and so that the edges of failure would fall inside the characterized region where possible. Each characterization range is between 1.6 and 2.5 times the width of the corresponding normal operating range. The widest expansion is culture pH, which was studied over 6.6 to 7.1 against a normal operating range of 6.75 to 6.95. The ranges are given in full in Table 4.1.

3 Materials and methods

3.1 Scale-down model and its qualification

The studies were executed on bench scale stirred tank bioreactors qualified as a model of the commercial vessel under SOP-1001. The scale-independent parameters, culture pH, temperature, dissolved oxygen, inoculation density and culture duration, were operated at the values proposed for commercial operation. The scale-dependent parameters, agitation, gas flow rate and headspace pressure, were set so that the small-scale vessel matched the commercial vessel on mixing time, on the volumetric oxygen mass transfer coefficient and on the rate at which carbon dioxide is stripped from the broth. A sparged bubble rises through a much shorter liquid column at bench scale than in the commercial vessel, so it leaves the broth carrying less carbon dioxide and the same specific gas flow strips less of it. Dissolved carbon dioxide was therefore held at its intended concentration by the composition of the sparge gas rather than by the gas flow rate.

Qualification compared the model against commercial-scale data on the input attributes of the culture and on the output attributes of the harvested pool. The acceptance criteria and the qualification record are held in PCP-003, and the model was released for characterization before the first design was executed. Within each design, the replicated centre points give the pure error term against which lack of fit is judged, and their reproducibility is reported in §5.1.

The model reproduces the commercial process at its set-point. The design space in §6 and the ranges in §7 rest on the further assumption that the parameter effects measured on the model act in the same direction and with the same size at commercial scale. That assumption is tested during Stage 2 and is not tested here.

3.2 Cell line, media and culture operation

The production culture is a recombinant CHO line expressing A-Mab, inoculated from the N-1 seed bioreactor into a chemically defined basal medium and operated in fed-batch mode under SOP-2003. A single nutrient feed is delivered during the run from the feed medium prepared under SOP-2004. Culture pH is controlled by carbon dioxide and base addition, dissolved oxygen by sparged air and oxygen, and temperature by the vessel jacket. Osmolality is set by the base and feed additions and by the accumulation of metabolites, and it is measured offline each day. The culture is harvested at the end of the run and clarified as described in PCR-004.

The nominal profile of the step at set-point operation is shown in Figure 3.1. Viable cell density rises to a peak of 13.5 million cells per mL and viability falls to 67 % by the end of the run, at which point the titer is 4.37 g/L. The decline in viability over the last days of the culture is what releases host cell protein and DNA into the broth, so a culture harvested a day later carries more of both into harvest.

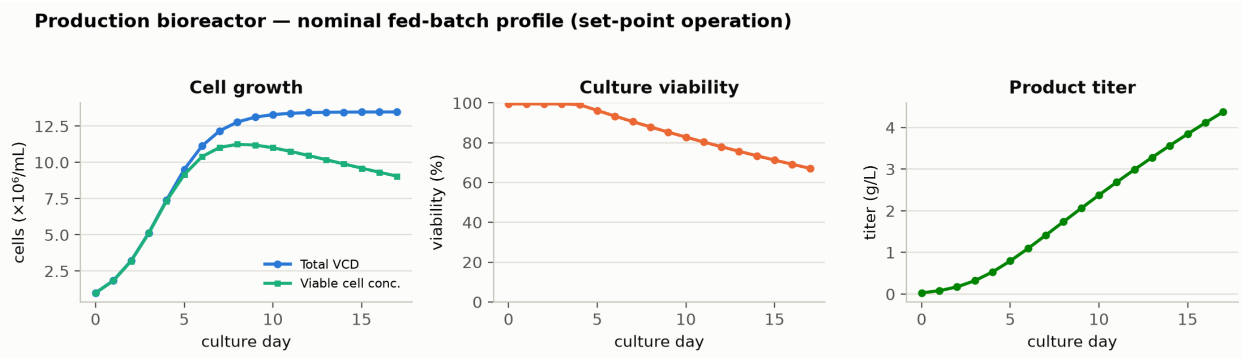

Figure 3.1: Nominal production bioreactor profile at set-point operation. Viable cell density and viability fall in the late phase of the culture, and the titer accumulates over the whole run.

The raw materials of the step are controlled to platform specification. The basal medium, the feed medium, the base and the gases are released against their material specifications and are identity and quality controlled at receipt. They were not varied in this study.

3.3 Analytical methods

The attributes of the harvested pool were measured with validated methods. The three glycan attributes are read from a single released N-glycan map by 2-AB HILIC-UPLC under SOP-3010 and AMV-3010, so afucosylation, galactosylation and high mannose share one analytical variance. Aggregate is measured by SEC-HPLC under SOP-3011 and AMV-3011, acidic charge variants by icIEF under SOP-3013 and AMV-3013, host cell protein by ELISA under SOP-3012 and AMV-3012, and residual DNA by qPCR under SOP-3014 and AMV-3014. Table 3.1 lists the controlled documents that govern the step and its methods, and Appendix D maps each attribute to its method and reportable unit.

Table 3.1: Controlled documents governing the execution and the analysis of this study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP

Reference	Title	Type
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

Two of the methods carry a validation record that bears directly on the interpretation of the results. The glycan map has an intermediate precision of 2.5 %RSD and a limit of quantitation of 0.5 % glycan, and about 30 % of the total observed variance in the glycan responses is attributable to the assay. The host cell protein ELISA has an intermediate precision of 9.5 %RSD and a limit of quantitation of 1 ng/mg, and about 40 % of the observed variance in that assay is analytical. The assay share of the variance matters for the centre-point reproducibility reported in §5.1, because a response whose analytical variance is high reproduces less well without the process being less reproducible.

3.4 Sampling plan

Each experimental run is one culture. In-process samples were drawn daily for viable cell density, viability, metabolites and osmolality, and the offline reference measurements of pH and dissolved carbon dioxide were taken at the same time. The quality attributes were measured once per run, on the clarified harvest pool, because the attribute of interest is the integral of what the culture secreted and not its value on any single day.

Replicated centre points carry the reproducibility of the whole system, from cell thaw to released result. The screening design carries 3 centre points and the response-surface design carries 4. Runs were executed in randomized order within each design and were assigned across the available scale-down bioreactors so that no factor level was confounded with a vessel.

3.5 Statistical methods

Every factor was analysed on a coded scale, with the set-point at 0 and the edges of the characterization range at −1 and +1. The screening data were fitted by ordinary least squares with all main effects and all two-factor interactions, and an effect is reported as twice the coded coefficient, which is the change in the response across the full range of the factor. The response-surface data were fitted with the full quadratic model in the four response-surface factors, comprising the main effects, the two-factor interactions and the pure quadratic terms. Significance is judged at $\alpha = 0.05$ throughout.

Model adequacy was assessed on four grounds. R^2 and adjusted R^2 report how much of the observed variation the model describes. Predicted R^2, computed from the PRESS statistic by leaving each run out in turn, reports how much of a new observation the model would describe. The overall F test reports whether the model explains more than the residual, and the ANOVA partition of the residual into lack of fit and pure error reports whether the deviation from the model is larger than the reproducibility of the replicated centre points. A factor with no significant term in any response is reported as having no demonstrated effect over the range studied.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches, with each parameter drawn inside its normal operating range and the fitted models used to predict the attributes of every batch. Capability is reported as a one-sided Cpk against the acceptance limit that binds. Proven acceptable ranges were computed from the response-surface models by two analyses, which are described with the results in §7.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 gives every parameter of the step, its set-point, its normal operating range, the range over which it was characterized, the study type and the classification that this report assigns to it in §9. The characterization range is the knowledge space of the step. The proven acceptable ranges are computed per attribute and parameter in §7.

Table 4.1: Production bioreactor parameters, ranges, study type and final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Culture pH	pH	6.85	6.75–6.95	6.6–7.1	WC-CPP	multivariate
Culture temperature	°C	35	34.5–35.5	34–36	WC-CPP	multivariate
Dissolved CO2 (pCO2)	mmHg	100	70–130	40–160	WC-CPP	multivariate
Osmolality	mOsm	400	375–425	360–440	WC-CPP	multivariate
Culture duration	days	17	16–18	15–19	WC-CPP	multivariate
Dissolved oxygen	% sat	50	40–60	30–70	KPP	univariate
Initial viable cell conc.	e6 cells/mL	1	0.8–1.2	0.5–1.5	KPP	univariate
Basal medium concentration	X	1.2	1–1.4	0.8–1.6	GPP	univariate
Nutrient feed-1 volume	% WV	12	10–14	9–15	KPP	univariate

The five parameters studied multivariately are coded A to E throughout this report and in the appendices. Table 4.2 gives the code, the parameter and the design each was studied in.

Table 4.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Culture pH	pH	screening + RSM
B	Culture temperature	°C	screening + RSM
C	Dissolved CO2 (pCO2)	mmHg	screening + RSM
D	Osmolality	mOsm	screening
E	Culture duration	days	screening + RSM

4.2 Screening design

The screening design is a half fraction of the two-level full factorial in the five multivariate factors, augmented with 3 centre points, for 19 runs in total. The fifth factor is assigned to the four-factor interaction, so the design is of resolution V. Every main effect and every two-factor interaction is therefore estimated free of the others, and the two-factor interactions are aliased only with three-factor and higher interactions, which are assumed negligible. The design matrix and the measured responses are given in Appendix A.

The fitted screening model carries 16 terms against 19 runs, so 3 degrees of freedom remain for the residual and they come from the centre point replicates. The design estimates every main effect and every two-factor interaction on all five responses at a cost of 19 cultures, and its predicted R^2 is negative for all five (§5.2). The design space in §6, the ranges in §7 and the capability in §8 are computed from the response-surface models described next.

4.3 Response-surface design

The response-surface design is a face-centred central composite design in the four factors that carried the largest effects in screening, namely culture pH, culture temperature, culture duration and dissolved carbon dioxide. It comprises 16 factorial runs, 8 axial runs and 4 centre points, for 28 runs in total. The axial runs sit on the faces of the cube rather than outside it, so no run was executed at a setting beyond the characterization ranges in Table 4.1. The design matrix and the measured responses are given in Appendix B.

Osmolality was not carried into the response-surface design. Summed across the five responses its screening effects came to 6.5, the smallest of the five factors, against 15.2 for dissolved carbon dioxide and 12.6 for culture pH. It was held at its set-point of 400 mOsm through the response-surface runs. Its two significant screening effects are carried forward as bounding estimates in §5.2, and the response-surface model carries no osmolality term. The design space in §6 therefore covers four parameters, and osmolality is held to 375 to 425 mOsm, a range 1.6 times narrower than the range screened.

The fitted response-surface model carries 15 terms against 28 runs, leaving 13 degrees of freedom for the residual, of which 3 are pure error from the replicated centre points. This is the model used for the design space, the proven acceptable ranges and the capability estimate.

4.4 Univariate assessment

Dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed-1 volume were assessed one at a time. Each was operated across the characterization range given in Table 4.1 with every other parameter at its set-point, and the harvest pool was tested against the full attribute panel. These four parameters were held out of the multivariate design for two different reasons. Dissolved oxygen is held by an independent sparge and agitation cascade, and a lower dissolved oxygen concentration leaves the cell less ATP for the synthesis of the nucleotide sugar donors, which moves the glycan fractions by less over the range assessed than any of the five multivariate parameters does. Initial viable cell concentration, basal medium concentration and nutrient feed-1 volume raise the titer without changing the glycans, because the feed replenishes the same amino acids and nucleotide sugar precursors through the run and the culture is fed to excess within the range studied.

The univariate assessment supports the classification of these four parameters in §9 and their ranges in Table 4.1. Interactions between them and the five multivariate factors were not tested.

5 Results

5.1 Centre-point performance and reproducibility

The replicated centre points measure the reproducibility of the whole system at the set-point, and they are the pure error term for every lack of fit test in this report. Table 5.1 and Table 5.2 give the mean, the standard deviation and the coefficient of variation of each response over the centre points of the two designs.

Table 5.1: Centre-point reproducibility in the screening design.

Response	n	Mean	SD	%CV
Afucosylation	3	7.93	0.502	6.33
Galactosylation	3	26.9	0.484	1.8
High mannose	3	6.28	0.721	11.5
Acidic variants	3	21.3	1.09	5.1
Aggregates (HMW)	3	1.59	0.0753	4.74

Table 5.2: Centre-point reproducibility in the response-surface design.

Response	n	Mean	SD	%CV
Afucosylation	4	7.61	0.355	4.66
Galactosylation	4	25.9	2.34	9.02
High mannose	4	5.66	0.559	9.88
Acidic variants	4	21.5	0.877	4.08
Aggregates (HMW)	4	1.51	0.0985	6.53

Acidic charge variants and afucosylation reproduce best, at 4.1 % and 4.7 % in the response-surface design. High mannose reproduces least well in both designs, at 11.5 % in screening and 9.9 % in the response-surface set. High mannose is the smallest of the three glycan populations at this set-point, with a centre-point mean of 5.7 % of glycans, so the same absolute analytical variation on the glycan map is a larger relative variation for it than for the others.

Galactosylation is the one response whose reproducibility differs between the two designs. It reproduced to 1.8 % over the screening centre points and to 9.0 % over the response-surface centre points. The larger spread in the response-surface set enters the pure error term directly, with a pure error mean square of 5.46 against a lack of fit mean square of 0.77. The lack of fit F ratio for galactosylation divides by that pure error mean square, and it is the smallest of the five (§5.3).

5.2 Screening: factor effects

The screening analysis identifies which parameters and which interactions move each response over the ranges studied. Table 5.3 summarizes the fit of the five screening models. R^2 is high for every response because the model is close to saturated, and the predicted R^2 values are negative for the same reason. A negative predicted R^2 means the model would predict a new run worse than the mean of the data would.

Table 5.3: Screening model summary for the five responses.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Afucosylation	19	0.965	0.79	-47.4	5.51	0.0926	1.06
Galactosylation	19	0.998	0.988	-1.33	103	0.00138	0.773
High mannose	19	0.942	0.65	-7.99	3.23	0.182	0.619
Acidic variants	19	0.994	0.963	-0.442	32.2	0.00766	0.957
Aggregates (HMW)	19	0.995	0.972	-2.44	42.2	0.00515	0.0829

Afucosylation was impacted by culture duration and by the interaction between culture pH and culture duration (Table 5.4). The interaction is the largest single effect estimate on the response at +2.46 ($p = 0.019$), and the culture duration main effect is -2.37 percentage points across the range studied ($p = 0.021$). The interaction between culture pH and dissolved carbon dioxide is also significant at -1.76. The culture pH main effect of -1.57 is not significant at $\alpha = 0.05$ ($p = 0.060$). The overall screening model for afucosylation is not significant either ($F = 5.51$, $p = 0.093$) on 3 residual degrees of freedom.

Table 5.4: Six largest screening effects on afucosylation.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AE	2.46	1.23	0.266	4.62	0.0191	*
E	-2.37	-1.19	0.266	-4.46	0.0209	*
AC	-1.76	-0.88	0.266	-3.31	0.0455	*
A	-1.57	-0.785	0.266	-2.95	0.06	
B	-1.34	-0.671	0.266	-2.53	0.0858	
D	-1.33	-0.664	0.266	-2.5	0.0878	

Galactosylation is the most strongly affected response of the five (Table 5.5). Culture duration reduces galactosylation by 7.82 percentage points across the range studied ($p < 0.001$), which is the largest main effect measured anywhere in this study. The interaction between culture pH and culture duration is +6.90, dissolved carbon dioxide is -5.79 and culture pH is -4.56. Of the 15 terms in the model, 12 are significant at $\alpha = 0.05$, including the osmolality main effect of -2.45 ($p = 0.008$).

Table 5.5: Six largest screening effects on galactosylation.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
E	-7.82	-3.91	0.193	-20.2	0.000264	***
AE	6.9	3.45	0.193	17.9	0.000382	***
C	-5.79	-2.9	0.193	-15	0.000644	***
A	-4.56	-2.28	0.193	-11.8	0.00131	**
BD	3.7	1.85	0.193	9.57	0.00242	**
AC	3.6	1.8	0.193	9.32	0.00261	**

Acidic charge variants were impacted mainly by dissolved carbon dioxide (Table 5.6). Its effect of -8.15 percentage points ($p < 0.001$) is twice the next largest, which is culture pH at -4.17. Culture temperature raises the response by +2.97 and osmolality lowers it by 2.55. The interaction between culture pH and culture temperature is significant at +1.98. Culture duration has no demonstrated effect on this response over the range studied (effect +0.023, $p = 0.964$). Time at culture temperature is a term in the deamidation rate, so an effect of culture duration was expected (§5.4).

Table 5.6: Six largest screening effects on acidic charge variants.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	-8.15	-4.08	0.239	-17	0.000441	***
A	-4.17	-2.08	0.239	-8.7	0.00319	**
B	2.97	1.48	0.239	6.19	0.00848	**
D	-2.55	-1.28	0.239	-5.34	0.0129	*
AB	1.98	0.992	0.239	4.15	0.0255	*
AC	-1.93	-0.964	0.239	-4.03	0.0275	*

High mannose and aggregate are the two responses with few active terms, and the full effect tables are in Appendix C. High mannose was impacted by culture pH (+1.51, $p = 0.016$) and by culture temperature (-1.39, $p = 0.021$), and no other term reaches significance. Aggregate was impacted by culture pH (+0.834) and by culture duration (+0.578), both at $p < 0.001$, and by nothing else of practical size. A higher culture pH brings the antibody nearer its isoelectric point, where the net charge on the molecule and the electrostatic repulsion between molecules are lowest, and a longer culture leaves the secreted product at culture temperature for longer.

Two conclusions were drawn from the screening stage and carried into the response-surface design. Culture pH, culture temperature, culture duration and dissolved carbon dioxide are active across the responses that matter most, and the interactions among them are large enough that a univariate treatment of any of the four would misstate its effect. Osmolality is active on two responses and smaller than the other four on every response, and it was held at its set-point in the second stage for the reason given in §4.3.

5.3 Response-surface models

The response-surface design was executed on the four factors carried forward, and the full quadratic model fitted to each response is the model from which §6, §7 and §8 are computed. Table 5.7 summarizes the fit of the five models.

Table 5.7: Response-surface model summary for the five responses.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Afucosylation	28	0.963	0.923	0.795	24.3	5.05e-07	0.584
Galactosylation	28	0.971	0.94	0.915	31.3	1.06e-07	1.36
High mannose	28	0.911	0.815	0.65	9.48	0.000118	0.472
Acidic variants	28	0.938	0.872	0.598	14.1	1.27e-05	1.57
Aggregates (HMW)	28	0.969	0.936	0.835	29.3	1.58e-07	0.0985

Every model is significant overall. R^2 runs from 0.911 for high mannose to 0.971 for galactosylation, and adjusted R^2 stays above 0.81 for all five. The predicted R^2 values are the more demanding measure, and they run from 0.60 for acidic charge variants to 0.92 for galactosylation. A model with a predicted R^2 of 0.60 describes a new observation less well than one at 0.92.

No response shows significant lack of fit. The two largest lack of fit F ratios are acidic charge variants at 3.85 ($p = 0.147$) and afucosylation at 3.23 ($p = 0.182$). Table 5.8 gives the full partition for acidic charge variants. The lack of fit mean square of 2.96 is about 3.9 times the pure error mean square, and the test does not reach significance with 3 degrees of freedom in the pure error term. The same test would reject the model at $\alpha = 0.20$.

Table 5.8: Analysis of variance and lack of fit for the acidic charge variant model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	484	14	34.6	14.1	1.27e-05
Residual	31.9	13	2.45	nan	nan
Lack of fit	29.6	10	2.96	3.85	0.147
Pure error	2.31	3	0.768	nan	nan

Table 5.9 gives the partition for galactosylation. Its pure error mean square of 5.46 is the largest of any response in this study, because the centre points of that design reproduced galactosylation to only 9.0 % (§5.1). That mean square is the denominator of the F ratio, which falls to 0.14, below the lack of fit mean square of 0.77. A test with a denominator that large would not separate a correct quadratic model from an incorrect one.

Table 5.9: Analysis of variance and lack of fit for the galactosylation model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	811	14	57.9	31.3	1.06e-07
Residual	24.1	13	1.85	nan	nan
Lack of fit	7.69	10	0.769	0.141	0.992
Pure error	16.4	3	5.46	nan	nan

Table 5.10 and Table 5.11 give the coefficient tables for the two glycan responses that drive the operating region. The coefficients are on the coded scale, so a main-effect coefficient is the change in the response for a move from the set-point to the edge of the characterization range, and an interaction coefficient is the change in one factor's slope for the same move in its partner. The coefficient tables for high mannose, acidic charge variants and aggregate are in Appendix C.

Table 5.10: Response-surface coefficients for afucosylation.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	8.04	0.202	39.8	5.74e-15	***
A	-0.997	0.138	-7.24	6.52e-06	***
B	-0.551	0.138	-4	0.00151	**
E	-1.3	0.138	-9.45	3.45e-07	***
C	-0.367	0.138	-2.67	0.0194	*
AB	-0.63	0.146	-4.32	0.000837	***
AE	1.32	0.146	9.06	5.6e-07	***
AC	-0.907	0.146	-6.21	3.16e-05	***
BE	0.362	0.146	2.48	0.0277	*
BC	-0.0703	0.146	-0.481	0.638	
EC	0.0322	0.146	0.221	0.829	
A^2	-1.37	0.364	-3.77	0.00233	**
B^2	-0.157	0.364	-0.431	0.673	
E^2	0.257	0.364	0.705	0.493	
C^2	0.117	0.364	0.323	0.752	

Table 5.11: Response-surface coefficients for galactosylation.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	25.8	0.47	54.8	9.11e-17	***
A	-2.21	0.321	-6.89	1.09e-05	***
B	2.3	0.321	7.18	7.13e-06	***
E	-3.82	0.321	-11.9	2.27e-08	***
C	-2.43	0.321	-7.56	4.11e-06	***
AB	1.67	0.34	4.92	0.000279	***
AE	2.72	0.34	7.99	2.28e-06	***

Term	Coef.	Std. err.	t	p-value	Sig.
AC	2.35	0.34	6.9	1.09e-05	***
BE	-0.217	0.34	-0.639	0.534	
BC	0.315	0.34	0.928	0.37	
EC	0.332	0.34	0.975	0.347	
A^2	-0.837	0.847	-0.988	0.341	
B^2	0.378	0.847	0.446	0.663	
E^2	0.937	0.847	1.11	0.289	
C^2	-0.373	0.847	-0.441	0.667	

For afucosylation the culture duration coefficient is -1.30 ($p < 0.001$), the culture pH coefficient is -1.00 ($p < 0.001$) and the interaction between them is +1.32 ($p < 0.001$). Afucosylation is also the only response with a significant quadratic term, -1.37 in culture pH ($p = 0.002$), so its surface is curved in the pH direction and falls away on both sides of the set-point. For galactosylation the four main effects are all significant, the largest being culture duration at -3.82, and 3 of the 6 interactions are significant, the largest being culture pH with culture duration at +2.72. No quadratic term is significant for galactosylation, so its surface over the four factors is a plane twisted by the interactions.

The remaining three responses are simpler. High mannose is governed by culture pH (+1.09, $p < 0.001$) with a smaller contribution from culture temperature. Acidic charge variants are governed by dissolved carbon dioxide (-4.09, $p < 0.001$), with culture pH and culture temperature contributing about half as much each and culture duration contributing nothing (-0.12, $p = 0.743$). Aggregate is governed by culture pH (+0.367) and culture duration (+0.281), both at $p < 0.001$, with a small significant quadratic term in culture duration of +0.143 ($p = 0.037$).

Figure 5.1 shows the five fitted surfaces over culture pH and culture duration with the other two factors at their set-points, and Figure 5.2 shows them over dissolved carbon dioxide and culture temperature. In the pH and duration plane the afucosylation and galactosylation contours run diagonally rather than parallel to either axis, and the galactosylation surface changes by more than 8 percentage points across the culture duration axis alone. In the second plane the acidic charge variant panel is the one that moves, and it does so almost entirely along the dissolved carbon dioxide axis.

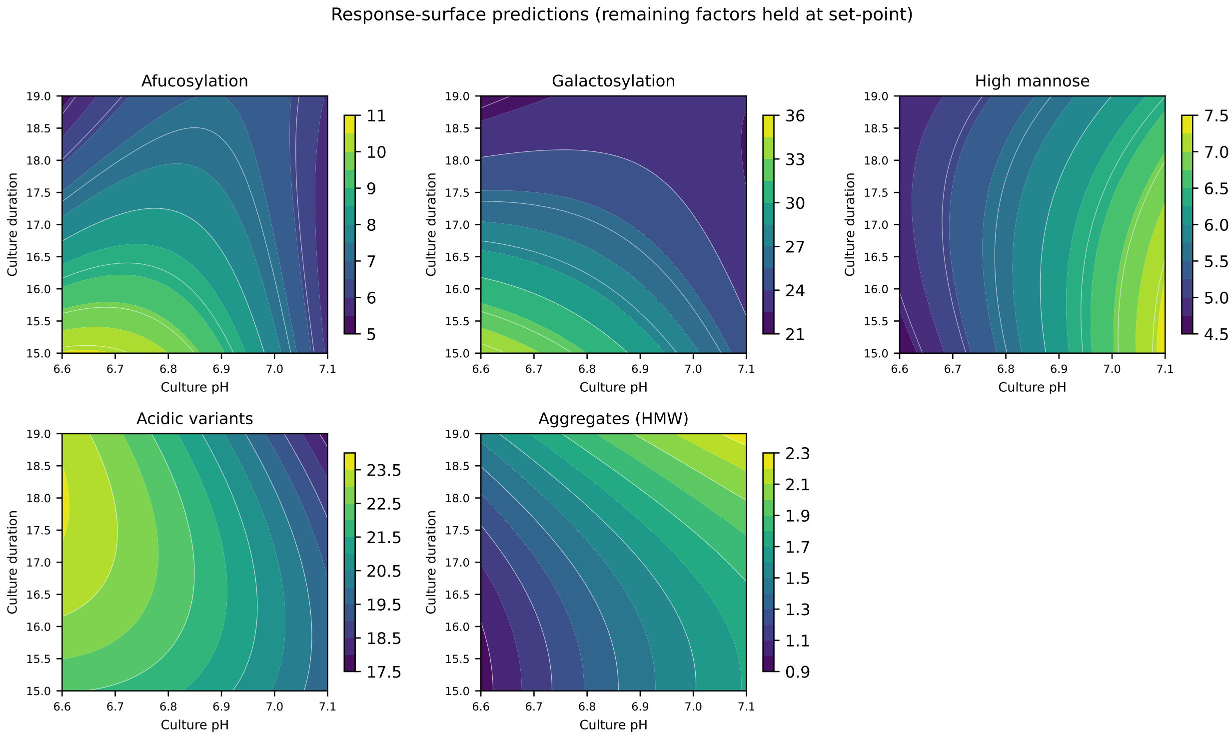

Figure 5.1: Fitted response surfaces over culture pH and culture duration, with culture temperature and dissolved carbon dioxide at their set-points.

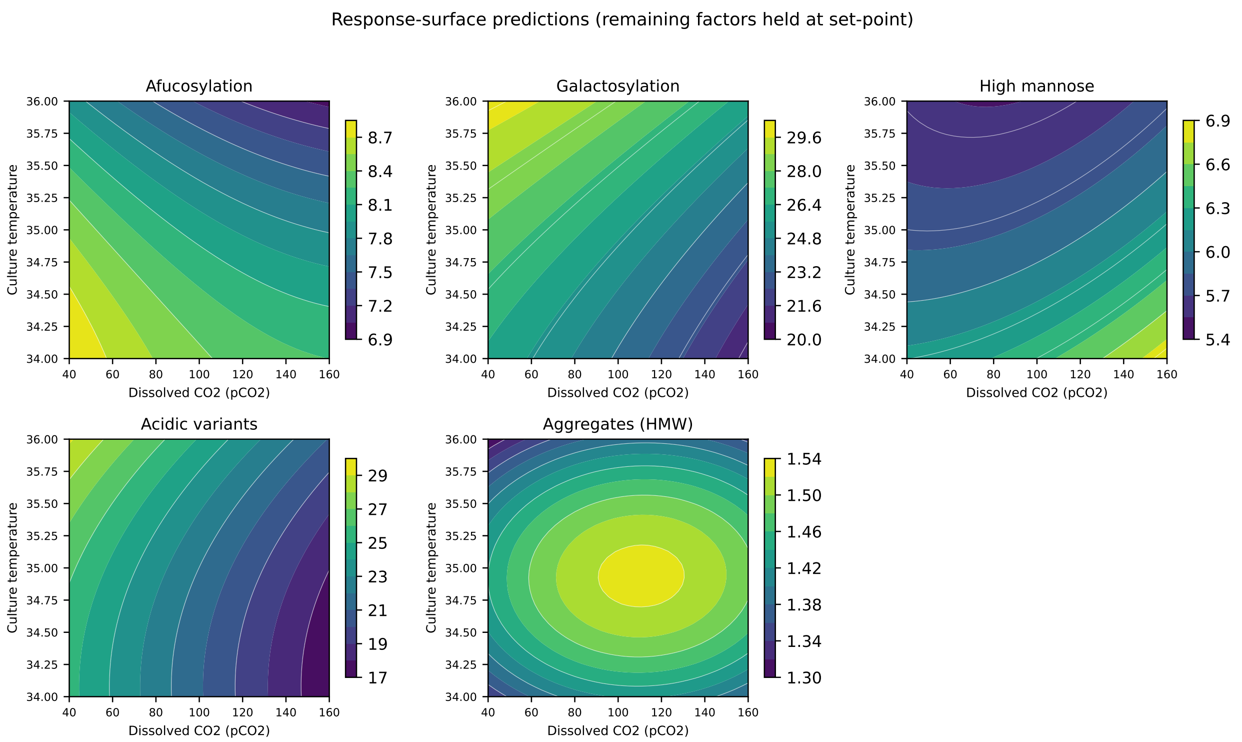

Figure 5.2: Fitted response surfaces over dissolved carbon dioxide and culture temperature, with culture pH and culture duration at their set-points.

The residual diagnostics for the two responses that carry the operating region are shown in Figure 5.3 and Figure 5.4. For afucosylation the residuals are scattered without structure against the predicted value, the normal quantile plot is close to linear, and the actual values track the predicted values across the whole range of the response. For acidic charge variants the residuals are larger, in proportion to the residual mean square of 2.45 in Table 5.8. Neither response shows a residual pattern that would indicate a missing term.

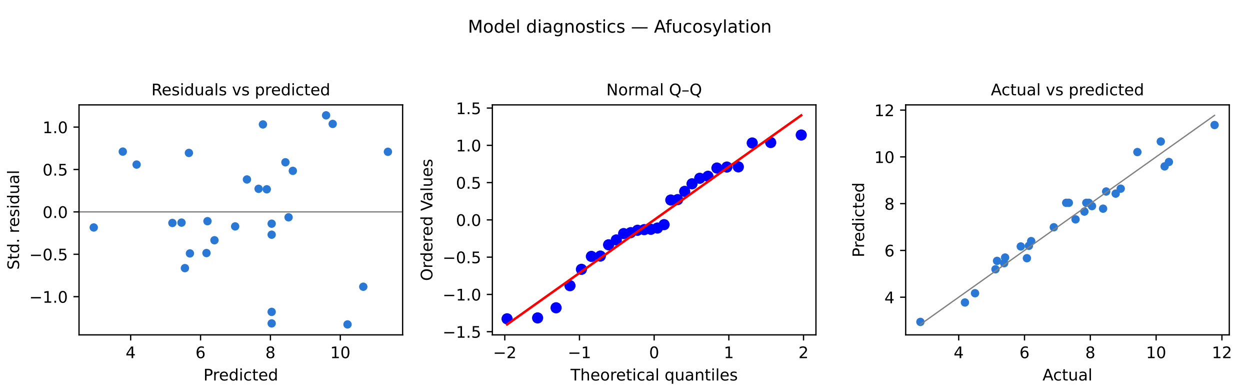

Figure 5.3: Residual diagnostics for the afucosylation response-surface model.

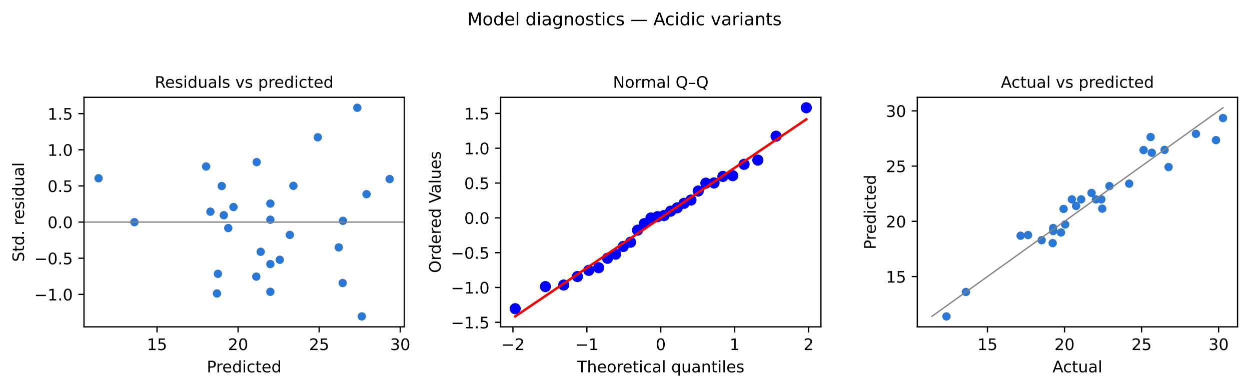

Figure 5.4: Residual diagnostics for the acidic charge variant response-surface model.

5.4 Mechanistic interpretation

The interaction between culture pH and culture duration on the two glycan attributes is the central result of the study. Culture pH sets how fast the galactosyltransferase and the fucosyltransferase work, through the pH of the Golgi lumen in which both enzymes sit. Culture duration sets how much of the harvested pool those enzymes made after their UDP-galactose and GDP-fucose donors had been drawn down. For afucosylation the slope on culture pH is -2.32 per coded unit at the short end of the culture duration range and +0.33 at the long end. For galactosylation the same slope moves from -4.93 to +0.51. A rise in culture pH therefore lowers both attributes in a short culture and leaves both almost unchanged in a long one. In a long culture the depleted donor pool limits the transfer, and raising the pH of the lumen no longer changes how much glycan the enzymes complete.

Read the other way, the same coefficients say that one day less culture raises galactosylation by 3.3 percentage points at the low end of the pH range and by 0.6 percentage points at the high end. A batch harvested a day early therefore leaves the reactor with a different glycan distribution depending on the pH at which it was grown, and the two parameters are controlled together rather than independently.

Dissolved carbon dioxide lowers the acidic charge variant fraction by 4.09 per coded unit, and culture pH lowers it by 2.01. Asparagine deamidation is base catalysed and runs faster as pH rises, so the deamidated species alone would give culture pH the opposite sign. The measured fraction also carries sialylated glycans and glycated lysines, and dissolved carbon dioxide lowers cytosolic pH and with it the rate at which the sialyltransferase adds sialic acid. The three contributions were not measured separately in this study, and the coefficients are reported as the net movement of their sum. Culture temperature raises the fraction by 1.90, which is the direction the deamidation rate takes as temperature rises.

High mannose rises with culture pH at +1.09 per coded unit and falls with culture temperature at -0.42. A less acidic Golgi lumen slows the mannosidases that trim the precursor glycan and the GlcNAc transferase that acts after them, so more antibody is secreted still carrying five or more mannose residues. A warmer culture runs those same enzymes faster and fewer such glycans escape. Culture duration has no significant effect on high mannose (-0.19, $p = 0.115$).

Aggregate forms in the broth rather than in the cell. It rises with culture pH at +0.367 per coded unit and with culture duration at +0.281, and no interaction between the two reaches significance (Table 13.11). A longer culture leaves more secreted antibody at culture temperature for longer, and a higher culture pH brings the molecule nearer its isoelectric point, where the net charge on the antibody is lowest and the electrostatic repulsion between two molecules is weakest. Both effects are small against the drug substance limit of 5 %.

One result was not predicted. Culture duration has no demonstrated effect on acidic charge variants over the range studied, with a coefficient of -0.12 and $p = 0.743$. Time at culture temperature is a term in the deamidation rate, so a longer culture was expected to raise the deamidated fraction. The proven acceptable range of culture duration for this attribute in Table 7.1 is the whole characterized range of 15 to 19 days.

6 Design space

The design space of this step is the multivariate region of culture pH, culture temperature, culture duration and dissolved carbon dioxide within which every quality attribute governed here is predicted to meet the criterion that applies to it. It is bounded by the characterization ranges in Table 4.1, because a prediction outside those ranges extrapolates the model. It is the subset of that knowledge space known to give acceptable attribute values, and the normal operating range is a smaller region inside it (International Council for Harmonisation 2009).

The region was evaluated by predicting all five responses on an evenly spaced grid of 14,641 points across the four dimensional coded cube and testing each point against every criterion at once. Across that grid, 57 % of the characterized region meets all five criteria simultaneously. Table 6.1 gives the share of the region that each response rejects on its own. Galactosylation rejects the largest share at 14 % and high mannose the smallest at 5 %, so galactosylation is the attribute that binds the operating region of this step.

Table 6.1: Share of the characterized region rejected by each response on its own, with the criterion applied.

Response	Share of region rejected (%)	In-process criterion
Afucosylation	11.3	≤ 9.43
Galactosylation	14.3	≤ 30.1
High mannose	5.25	≤ 7.28
Acidic variants	13.1	≤ 25.8
Aggregates (HMW)	11.3	≤ 1.86

The principal plane of the design space is culture pH against culture duration (Figure 5.1). Those two parameters carry the largest main effects on the two binding attributes and the largest interaction between them, so the boundary of the acceptable region in that plane is a curve rather than a rectangle. Dissolved carbon dioxide bounds the region through acidic charge variants and, more weakly, through galactosylation. Culture temperature bounds it least.

The normal operating range sits inside the design space at its centre and reaches the boundary at 3 of its 16 corners. Each of the three combines the low end of the culture pH range with the shortest culture duration, the pair of settings at which a rise in pH lowers galactosylation and afucosylation most steeply (§5.4). At the corner that adds the low end of the temperature range and the high end of the dissolved carbon dioxide range, the model predicts afucosylation of 9.431 % against an in-process alert limit of 9.426 %. At the corner combining the low end of the pH range, the shortest culture, the high end of the temperature range and the low end of the dissolved carbon dioxide range, it predicts galactosylation of 31.8 % against an alert limit of 30.1 %. Every corner remains well inside the drug substance acceptance criteria of 2 to 13 % and 10 to 40 %. The alert limits and what they are derived from are described in §7, and the consequence for the control strategy is in §10.

Three bounds apply to this design space. It is defined on mean predictions from the fitted models, so a batch run exactly on its boundary meets the criterion about half the time if the variation around the predicted mean is symmetrical. The propagated ranges in §7 are computed from a predictive interval instead. It covers four of the five multivariate parameters, and osmolality is held at its set-point rather than modelled (§4.3). It is defined on the qualified scale-down model and is not confirmed at commercial scale at the edges of the ranges.

Movement within the design space is not a regulatory change (International Council for Harmonisation 2009). Every planned movement is still assessed prospectively under the pharmaceutical quality system before it is made, and the assessment records the risk of the particular move and the knowledge that supports it (International Council for Harmonisation 2008). Operation outside the characterized ranges is outside the knowledge space of this study and is handled as a deviation.

7 Proven acceptable ranges

A proven acceptable range is the range of a single parameter over which the attribute it governs is shown by supportive data to stay within its criterion. The design space in §6 is a multivariate statement and the proven acceptable ranges are the univariate statements that sit inside it, one per attribute and parameter. They are the form in which the ranges enter the batch record and the control strategy.

Each range in this section is measured against the in-process criterion that applies to this step, not against the drug substance acceptance criterion. For the four attributes that the purification train does not modify, the demonstrated commercial-scale capability of the step is the tighter yardstick, so the criterion is an alert limit set at the capability mean plus 2.5 standard deviations. For aggregate, which low pH inactivation raises and cation exchange then reduces, the criterion is carried back from the drug substance limit through the clearance the downstream steps deliver and divided by a factor of 6.8, so that the limit sits well below the concentration at which the downstream train could no longer bring the batch to specification. The criteria are listed in the fifth column of Table 7.1. Every one of them sits inside the corresponding drug substance criterion in Table 2.1.

Two analyses were run for each attribute and parameter, and both are reported. The first holds the other three response-surface parameters at their set-points, which at this step are also the centres of the characterization ranges. The second propagates normal operating range variation in the other parameters by Monte-Carlo simulation of the fitted model, adds the residual variation of the model, and takes the range over which the 95 % predictive interval stays within the criterion. The first analysis answers what one parameter may do when everything else is on target. The second answers what it may do in routine manufacture, and it is the range the control strategy uses.

Table 7.1: Proven acceptable ranges per attribute and parameter, at the set-point and with normal operating range variation propagated.

CQA	Parameter	Char. range	Unit	Criterion	PAR (set-point)	PAR (NOR)
Afucosylation	Culture pH	6.6–7.1	pH	≤ 9.43	6.6–7.1	6.82–7.1
Afucosylation	Culture temperature	34–36	°C	≤ 9.43	34–36	34.7–36
Afucosylation	Culture duration	15–19	days	≤ 9.43	15.2–19	16.8–19
Afucosylation	Dissolved CO2 (pCO2)	40–160	mmHg	≤ 9.43	40–160	82–160

CQA	Parameter	Char. range	Unit	Crite-rion	PAR (set-point)	PAR (NOR)
Galactosyla-tion	Culture pH	6.6–7.1	pH	≤ 30.1	6.6–7.1	6.74–7.1
Galactosyla-tion	Culture temperature	34–36	°C	≤ 30.1	34–36	34–35.5
Galactosyla-tion	Culture duration	15–19	days	≤ 30.1	15.2–19	16.4–19
Galactosyla-tion	Dissolved CO2 (pCO2)	40–160	mmHg	≤ 30.1	40–160	74.5–160
High mannose	Culture pH	6.6–7.1	pH	≤ 7.28	6.6–7.1	6.6–6.94
High mannose	Culture temperature	34–36	°C	≤ 7.28	34–36	34.1–36
High mannose	Culture duration	15–19	days	≤ 7.28	15–19	15–19
High mannose	Dissolved CO2 (pCO2)	40–160	mmHg	≤ 7.28	40–160	40–156
Acidic variants	Culture pH	6.6–7.1	pH	≤ 25.8	6.6–7.1	6.82–7.1
Acidic variants	Culture temperature	34–36	°C	≤ 25.8	34–36	34–35.2
Acidic variants	Culture duration	15–19	days	≤ 25.8	15–19	15–19
Acidic variants	Dissolved CO2 (pCO2)	40–160	mmHg	≤ 25.8	46.6–160	92.5–160
Aggregates (HMW)	Culture pH	6.6–7.1	pH	≤ 1.86	6.6–7.1	6.6–6.93
Aggregates (HMW)	Culture temperature	34–36	°C	≤ 1.86	34–36	34–36
Aggregates (HMW)	Culture duration	15–19	days	≤ 1.86	15–18.7	15–17.7
Aggregates (HMW)	Dissolved CO2 (pCO2)	40–160	mmHg	≤ 1.86	40–160	40–160

The set-point analysis covers the whole characterization range for most pairs in Table 7.1. The exceptions are culture duration for afucosylation and galactosylation, where the acceptable range starts at 15.2 days, culture duration for aggregate, where it ends at 18.7 days, and dissolved carbon dioxide for acidic charge variants, where it starts at 46.6 mmHg. Each of those 4 edges is an edge of failure of this step against its in-process criteria, and all of them fall inside the characterization ranges in Table 4.1.

The propagated analysis is never wider than the set-point analysis, and for several pairs it is narrower than the normal operating range itself. For afucosylation the propagated range for culture duration is 16.8–19 days against a normal operating range of 16 to 18 days, and the propagated range for culture pH is 6.82–7.1 against a normal operating range of 6.75 to 6.95. For galactosylation the propagated range for culture duration is 16.4–19 days. For acidic charge variants the propagated range for dissolved carbon dioxide is 92.5–160 mmHg against a normal operating range of 70 to 130 mmHg. In each case the lower part of the normal operating range is not covered.

The criterion behind those ranges is an alert limit set at 2.5 standard deviations of the demonstrated capability, and the propagated analysis asks that a 95 % predictive interval, which carries the analytical variance of the glycan map and the residual variance of the model, stays inside that limit while three other parameters vary across their full normal operating ranges at once. Every corner of the normal operating range meets the drug substance acceptance criteria (§6), and the capability against those criteria is reported in §8.

Figure 7.1 to Figure 7.5 show the analysis for each attribute against its governing parameter, which is the response-surface factor with the largest main effect on it. Each figure has the same two panels. The left panel is the set-point analysis and the right panel is the propagated analysis, with the median, the 95 % predictive interval, the acceptance limits, the set-point, the normal operating range and the acceptable region shaded green.

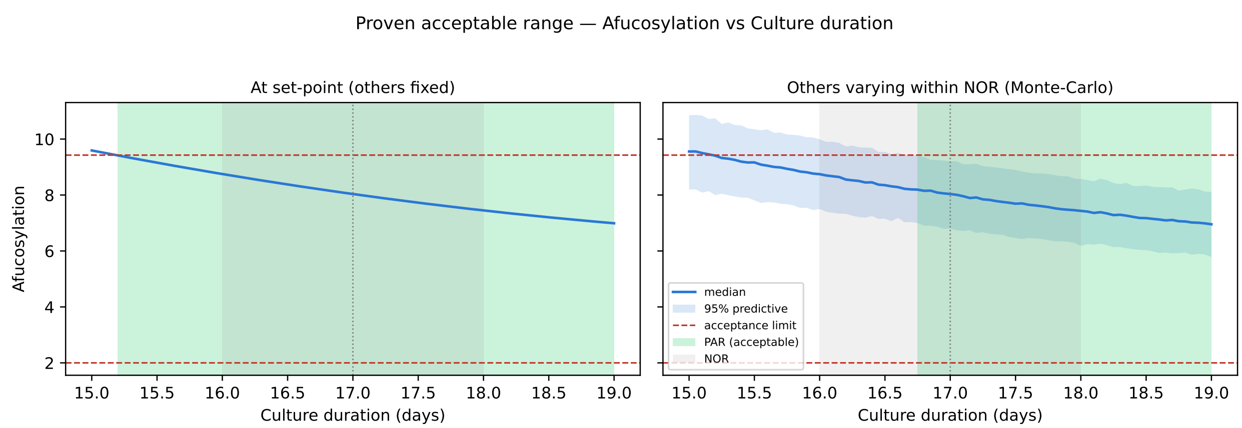

Figure 7.1: Proven acceptable range for afucosylation against culture duration, its governing parameter.

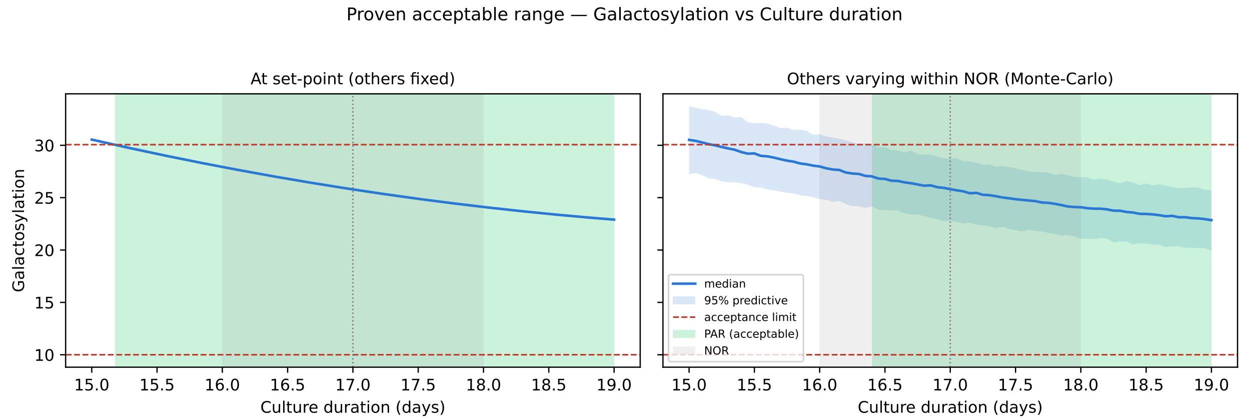

Figure 7.2: Proven acceptable range for galactosylation against culture duration, its governing parameter.

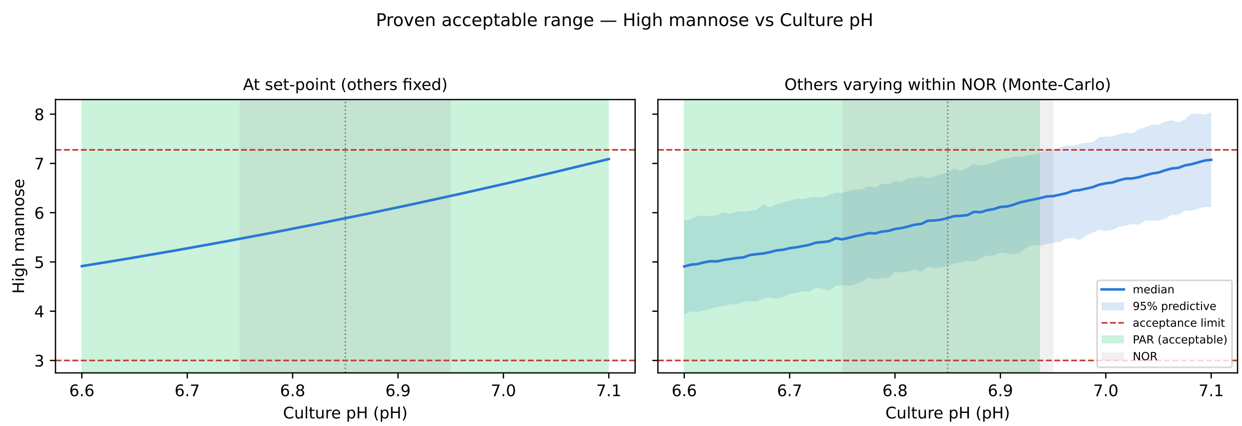

Figure 7.3: Proven acceptable range for high mannose against culture pH, its governing parameter.

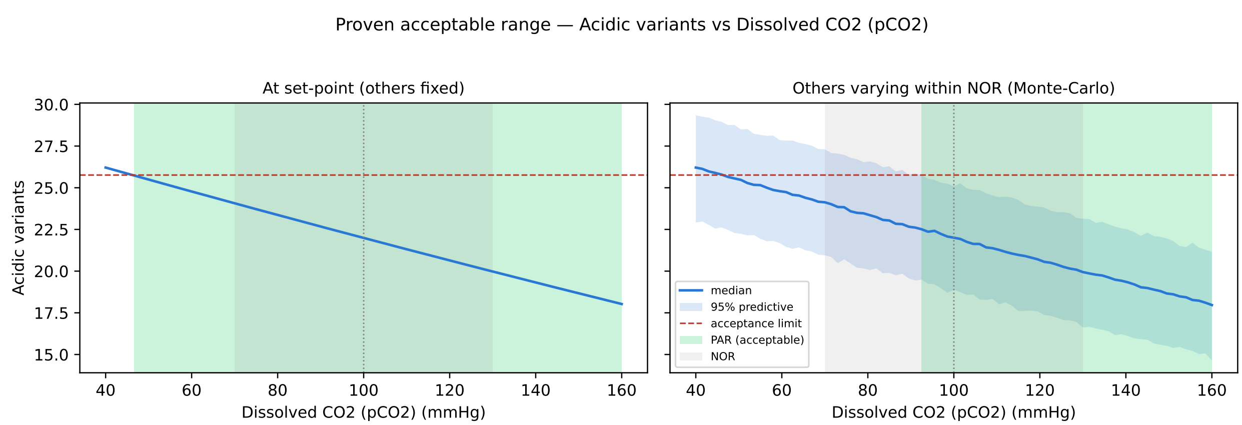

Figure 7.4: Proven acceptable range for acidic charge variants against dissolved carbon dioxide, its governing parameter.

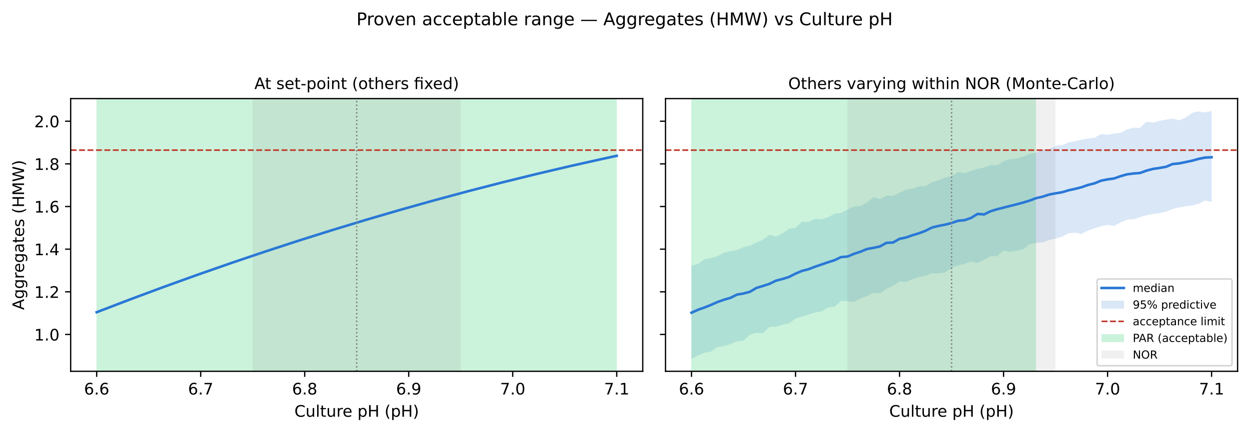

Figure 7.5: Proven acceptable range for aggregate against culture pH, its governing parameter.

The univariate ranges in Table 7.1 and the multivariate region in §6 are two views of the same models and they are used for different purposes. A parameter that stays inside its propagated range while the others are inside their normal operating ranges is inside the design space by construction. A parameter outside its propagated range is not necessarily outside the design space, because the design space is the intersection over all four parameters and a movement in one can be compensated by the position of another. Any such movement is assessed against the model rather than against the univariate range.

8 Process capability and robustness

Capability was estimated at commercial scale by simulating 2,000 batches. Each batch was generated by drawing every process parameter inside its normal operating range, predicting the attributes of that batch from the fitted models, and applying the clearance and the variability of the downstream steps where the attribute is one the train modifies. Table 8.1 gives the resulting distribution of each attribute against its drug substance acceptance criterion, with the one-sided Cpk against the limit that binds.

Table 8.1: Commercial-scale capability of the attributes set at this step, from the Monte-Carlo simulation.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Afucosylation	H	two_sided	2–13	7.98 ± 0.58	2.89
Galactosylation (total %Gal)	H	two_sided	10–40	26 ± 1.6	2.86
High Mannose	VH	two_sided	3–10	5.99 ± 0.51	1.94
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.1
Acidic charge variants (deamidation)	VL	upper	18–40	22.3 ± 1.4	4.26
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.14
Residual DNA	VL	upper	0–0.001	0.0001 ± 0	10.8

Every attribute meets its drug substance acceptance criterion across the simulated campaign. The tightest capability is high mannose, at a one-sided Cpk of 1.94. Its simulated mean of 5.99 % of glycans sits between acceptance limits of 3 and 10 %, and the lower limit is the one that binds, at 5.8 standard deviations from the mean. The two efficacy-linked glycan attributes follow, with afucosylation at 2.89 and galactosylation at 2.86, and in both cases the upper limit binds. Acidic charge variants reach 4.26.

No numerical acceptance criterion for Cpk is defined for this characterization campaign, and each value above is reported with the margin that produces it. The comparison across the whole train, including the attributes that other steps govern, is made in PCMR-001.

Three of the seven attributes in Table 8.1 are not set by this step alone. Aggregate reaches a Cpk of 16.1 at the drug substance because low pH viral inactivation raises it and cation exchange is the principal polishing step for it, and those contributions are characterized in PCR-006 and PCR-007. Host cell protein reaches 6.1 and residual DNA 10.8 because Protein A capture, cation exchange and anion exchange clear both by orders of magnitude, as reported in PCR-005, PCR-007 and PCR-008. What this step contributes to those three attributes is the load handed to the purification train, and that load rises with culture duration as viability falls over the last days of the run (Figure 3.1). The purification train does not change the four glycan and charge variant attributes, so the capability in Table 8.1 for those four is the capability of this step.

The capability figures above were produced with every parameter varying across its normal operating range at once, so they already carry the multivariate variation that the propagated ranges in §7 test against the tighter in-process alert limits. An attribute may reach an alert limit at a corner of the normal operating range while the batch remains far inside the drug substance criterion, and for afucosylation and galactosylation it does.

9 Parameter classification

Parameters were classified under SOP-4001 from their demonstrated effect on a quality attribute and from the capability with which they are controlled (CMC Biotech Working Group 2009). A parameter whose variation affects a critical quality attribute is a critical process parameter or a well controlled critical process parameter, and the two are separated by the risk of the parameter leaving the design space in routine operation. A parameter that affects process performance but no quality attribute is a key process parameter. A parameter that affects neither is a general process parameter. The classification of every parameter is in Table 4.1 and the reasoning is below.

- Culture pH is a well controlled critical process parameter. A rise in culture pH raises high mannose and aggregate, and it lowers galactosylation and afucosylation at the short end of the culture duration range while leaving both almost unchanged at the long end (§5.4). It is held to 6.75 to 6.95 by carbon dioxide and base addition under a calibrated probe loop, which is narrow against the characterized range of 6.6 to 7.1, so the risk of operating outside the design space is low.
- Culture temperature is a well controlled critical process parameter. A rise in culture temperature raises galactosylation and the acidic charge variant fraction and lowers afucosylation, and it is held to 34.5 to 35.5 °C by the vessel jacket, which is the most reliably controlled of the five multivariate parameters.
- Dissolved carbon dioxide is a well controlled critical process parameter. A rise in dissolved carbon dioxide lowers the acidic charge variant fraction, by the largest single coefficient measured in this study, and it lowers galactosylation. It is controlled indirectly, through sparge gas composition and flow, and its normal operating range of 70 to 130 mmHg is wide in absolute terms. It is nonetheless well controlled rather than critical, because the attribute it governs most strongly is of very low criticality and because the attribute that binds the operating region responds to it less than to culture duration.
- Osmolality is a well controlled critical process parameter. A rise in osmolality lowers galactosylation and the acidic charge variant fraction, both significantly in screening, and it was held at its set-point in the response-surface stage, so its effect is bounded by the screening estimate rather than modelled. It is controlled by the base and feed additions within 375 to 425 mOsm and is measured offline daily.
- Culture duration is a well controlled critical process parameter. A longer culture lowers galactosylation and afucosylation, by the largest main effects in the study, raises aggregate, and raises the host cell protein and DNA load carried into harvest. Harvest timing is a scheduled operation with a tolerance of hours

against a normal operating range of 16 to 18 days, so the parameter is controlled with certainty even though its effect is the largest of the five.

- Dissolved oxygen is a key process parameter. It is held by an independent cascade of sparging and agitation, it moves the glycan fractions only weakly within the range assessed, and it was not carried into the multivariate design. A concentration below the level at which respiration becomes oxygen limited lowers the specific growth rate and the titer, so the parameter is monitored on every batch.
- Initial viable cell concentration is a key process parameter. A higher inoculation density brings the culture to its peak viable cell concentration sooner, which raises the integral of viable cell concentration over the run and with it the titer. The specific productivity and the glycosylation of each cell are unchanged by the starting density, so the titer moves further than any quality attribute does.
- Nutrient feed-1 volume is a key process parameter. It replenishes glucose, amino acids and glycosylation precursors, and within the range assessed the culture is fed to excess, so the parameter raises the titer rather than the glycan fractions. Its delivery is verified gravimetrically before each run (§13.2).
- Basal medium concentration is a general process parameter. A higher basal medium concentration raises the starting supply of amino acids and nucleotide sugar precursors, the feed replenishes the same components through the run, and no effect on a quality attribute was demonstrated over the range assessed.

There is no critical process parameter at this step. Every parameter that is linked to a quality attribute is held inside a normal operating range that the equipment controls reliably and that sits inside the region characterized here, so all five quality-linked parameters were classified as well controlled critical process parameters.

10 Contribution to the control strategy

This step contributes five elements to the control strategy of the drug substance process.

The first is the set of parameter ranges in Table 4.1. The five quality-linked parameters are controlled to their normal operating ranges in the batch record, and the ranges are justified by the design space in §6 and the proven acceptable ranges in §7.

The second is in-process monitoring. Viable cell density, viability, metabolites and osmolality are measured daily, and culture pH and dissolved carbon dioxide are verified each day against an offline reference. Alert limits are applied to the four attributes the purification train does not modify, at the capability mean plus 2.5 standard deviations, and they are the limits used in §7. Their purpose is to detect a batch that is drifting inside its acceptance criteria, and a batch that reaches one is investigated rather than rejected.

The third is release and characterization testing. The attributes in Appendix D are measured on the harvest pool for every batch during Stage 2 and on the drug substance against the acceptance criteria in Table 2.1. The three glycan attributes are read from one glycan map, so a single released result carries all three.

The fourth is life-cycle control. The four attributes this step alone determines are trended under continued process verification, and culture duration is trended against the harvest host cell protein and DNA concentrations, which rise as viability falls over the last days of the run. Any change to the cell line, the medium or the feed is assessed against the models in this report before it is made.

The fifth is the two input controls that the deviations in §13 produced. The dissolved carbon dioxide probe is verified against an offline reference each day of the culture, and the nutrient feed pump is checked gravimetrically before each run.

The attributes this step forms are not all controlled here. Aggregate is reduced at cation exchange (PCR-007) after being raised at low pH viral inactivation (PCR-006). Host cell protein and residual DNA are cleared at Protein A capture (PCR-005), cation exchange (PCR-007) and anion exchange (PCR-008). This report claims control of the load those steps receive and of the four attributes that no downstream step changes. The consolidated picture is in PCMR-001.

11 Discussion

The step is now characterized over a region substantially wider than its routine control ranges, and the parameters that matter are identified with their directions and their sizes. A longer culture lowers galactosylation and afucosylation, and a rise in culture pH lowers both at the short end of the culture duration range and leaves both almost unchanged at the long end. A rise in dissolved carbon dioxide lowers the acidic charge variant fraction, and a rise in culture pH raises high mannose and aggregate. A rise in osmolality lowers galactosylation and the acidic charge variant fraction by the amounts the screening design estimated (Table 13.5, Table 13.7).

Confidence in the scale-down model rests on its qualification against commercial-scale data at the set-point and on the reproducibility of its centre points, which is reported in §5.1 and is between 4.1 and 9.9 % across the five responses in the response-surface design. Reproducibility of that order over a culture of 17 days, measured from cell thaw to released glycan map, is what makes effects of the size reported in §5.2 detectable at all.

Two deviations were recorded and both were retained (§13). Neither invalidated a run and neither changed a conclusion. The dissolved carbon dioxide probe offset moves the most sensitive response by less than half of that response's own centre-point standard deviation, and the nutrient feed under-delivery left the delivered volume inside the normal operating range of the parameter. Both produced a control that did not exist before, which is recorded in §10.

Four limitations bound what this report establishes.

The first is the scale-down model. Every result here is measured at small scale, and the design space is not confirmed at commercial scale at the edges of the ranges. The confirmation is a Stage 2 activity.

The second is the form of the model. The design space in §6 is built from mean predictions, so a batch run on its boundary meets the criterion about half the time if the variation around the predicted mean is symmetrical. The propagated ranges in §7 are computed from a 95 % predictive interval instead, and they are narrower than the multivariate region.

The third is the coverage of the response-surface model. It covers four of the five multivariate parameters. Osmolality was held at its set-point through the second stage, so its two significant effects are bounded by the screening estimates and are not modelled, and its interactions with the other four are not estimated beyond the two-factor terms in the screening model.

The fourth is the acidic charge variant model, which has the lowest predicted R^2 of the five at 0.60 and the largest lack of fit F ratio at 3.85. Its proven acceptable ranges are reported in Table 7.1 with the same method as the others, and they carry more uncertainty. The attribute is of very low criticality and its capability is 4.26, so the weaker model does not affect the operating region.

The univariate parameters were assessed one at a time. That approach cannot determine an interaction between one of them and a multivariate factor.

12 Conclusions

The production bioreactor has been characterized on a qualified scale-down model over ranges between 1.6 and 2.5 times the width of the routine control ranges. A multivariate operating region has been defined for culture pH, culture temperature, culture duration and dissolved carbon dioxide, and it covers 57 % of the characterized region, with galactosylation as the binding attribute.

Proven acceptable ranges have been established for every attribute and parameter pair of the response-surface model, at the set-point and with normal operating range variation propagated, and they are reported against the in-process criteria that apply to this step. The 4 edges of failure identified in §7 all fall inside the characterized region.

All 7 quality attributes set at this step meet their drug substance acceptance criteria at commercial scale in a simulation of 2,000 batches. The tightest capability is high mannose, at a one-sided Cpk of 1.94.

Of the 9 parameters, 5 were classified as well controlled critical process parameters, 3 as key process parameters and 1 as a general process parameter, and there is no critical process parameter at this step. Both deviations were retained without effect on the conclusions. The results support the Stage 2 ranges for this step and are rolled up into PCMR-001.

13 Deviations from the plan

The register in Table 13.1 lists the 2 deviations from PCP-003 recorded during execution. Both were investigated to root cause, both were dispositioned as retained, and neither invalidated a run or changed a conclusion of this report.

Table 13.1: Register of deviations recorded during execution of PCP-003.

Deviation	Summary	Detected during	Disposition
DEV-003-01	Dissolved-CO2 probe drift during screening run 4	in-process monitoring vs offline reference	retained
DEV-003-02	Nutrient feed-1 under-delivery on a centre-point batch	post-execution mass reconciliation	retained

13.1 DEV-003-01, dissolved carbon dioxide probe drift

During screening run 4 the in-process dissolved carbon dioxide reading differed from the offline reference by 6.5 mmHg. The discrepancy was found by the daily comparison of the in-process probe against the offline blood gas reference, which is part of the sampling plan in §3.4.

The investigation traced the discrepancy to drift of the probe on the scale-down bioreactor EQ-BRX-205 between scheduled calibrations. The probe was inside its calibration interval at execution, with the next calibration due 2026-06-14. The probe was recalibrated before the next run and the offline reference was retained as the reported value for the affected run.

The impact was assessed by propagating the offset through the screening models. An offset of 6.5 mmHg is 0.11 coded units on a characterization range of 40 to 160 mmHg. Applied to the most sensitive response, acidic charge variants, it moves the prediction by 0.44 percentage points, against a centre-point standard deviation of 1.09 for the same response in the same design. The three glycan responses move by no more than 0.31 percentage points. A shift smaller than the reproducibility of the measurement cannot change which terms the screening analysis identifies as active, and dissolved carbon dioxide was in any case carried into the response-surface design, where it was measured again over its full range with a calibrated probe.

The run was retained in the analysis and the deviation was closed as retained. The daily offline verification of the dissolved carbon dioxide probe was added to the control strategy for the step (§10).

13.2 DEV-003-02, nutrient feed-1 under-delivery

Post-execution mass reconciliation showed that one centre point batch of the screening design received 4.2 % less nutrient feed-1 than the batch record specified. The delivered volume was 11.5 % of working volume against a set-point of 12 % and a normal operating range of 10 to 14 %, so the batch was fed inside the normal operating range of the parameter but below its target.

The investigation traced the shortfall to a calibration offset on the feed pump. The feed lot LOT-FED-3120 was within its expiry of 2026-09-30 and its release record was in order. The pump was recalibrated and a gravimetric check of the delivered feed mass was introduced before each run.

Two arguments bound the impact. The first is mechanistic. Nutrient feed-1 volume is not a factor of either design and the culture is fed to excess across the range assessed. A shortfall that leaves the delivered volume inside the normal operating range leaves the glucose, the amino acids and the nucleotide sugar precursors above the concentrations at which they limit either growth or glycan processing.

The second is analytical. The affected batch is one of the 3 centre points of the screening design, which are the pure error term for that design, so an effect on it would inflate the pure error rather than bias an effect estimate. The released glycan map was re-tested, and a set of 3 determinations by AMV-3010 carries a relative 95 % confidence half width of 6.21 %. The spread of the three screening centre points is 1.8 % for galactosylation and 6.3 % for afucosylation, both inside that interval. High mannose spread further, at 11.5 %, but the centre points of the response-surface design, which were not affected by this deviation, reproduced that response to 9.9 %.

The batch was retained as a centre point and the deviation was closed as retained. The gravimetric feed check is part of the control strategy for the step (§10).

Appendix A. Screening design matrix and responses

Table 13.2 gives the executed screening design in coded units with the measured responses. The factor letters are those of Table 4.2.

Table 13.2: Screening design matrix in coded units, with the measured responses.

Run	Type	A	B	C	D	E	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
1	factorial	-1	-1	-1	-1	1	5.18	27.9	6.61	26.4	1.3
2	factorial	-1	-1	-1	1	-1	8.93	36.1	6.44	26.5	0.783
3	factorial	-1	-1	1	-1	-1	11.8	34.6	5.63	22.3	0.785
4	factorial	-1	-1	1	1	1	4.83	14.7	5.46	17.9	1.48
5	factorial	-1	1	-1	-1	-1	9.72	41.4	4.61	27	0.859
6	factorial	-1	1	-1	1	1	4.68	27.3	4.68	27.6	1.26
7	factorial	-1	1	1	-1	1	6.22	14.5	4.65	23.1	1.48
8	factorial	-1	1	1	1	-1	9.83	31.2	4.45	19.3	0.948
9	factorial	1	-1	-1	-1	-1	9.61	28	7.77	24.1	1.51
10	factorial	1	-1	-1	1	1	7.51	17	7.89	20.4	2.26

Run	Type	A	B	C	D	E	Afuco-sylation	Galacto-sylation	High man-nose	Acidic variants	Aggre-gates (HMW)
11	fac-to-rial	1	-1	1	-1	1	7.13	19.5	7.34	12.6	2.53
12	fac-to-rial	1	-1	1	1	-1	5.29	17.6	7.01	11.3	1.61
13	fac-to-rial	1	1	-1	-1	1	6.74	28.3	6.43	29.6	2.18
14	fac-to-rial	1	1	-1	1	-1	5.4	26.7	5.94	24.4	1.62
15	fac-to-rial	1	1	1	-1	-1	3.82	25.2	6.1	18.4	1.8
16	fac-to-rial	1	1	1	1	1	3.08	29.1	6.15	15.8	2.05
17	cen-ter	0	0	0	0	0	7.8	26.4	5.88	21.9	1.64
18	cen-ter	0	0	0	0	0	7.51	27.4	5.84	20	1.63
19	cen-ter	0	0	0	0	0	8.48	27	7.11	21.8	1.5

Appendix B. Response-surface design matrix and responses

Table 13.3 gives the executed face-centred central composite design in coded units with the measured responses. Osmolality was held at its set-point throughout and is therefore not a column of this matrix.

Table 13.3: Response-surface design matrix in coded units, with the measured responses.

Run	Type	A	B	E	C	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
1	factorial	-1	-1	-1	-1	9.43	38.8	5.44	29.8	0.827
2	factorial	-1	-1	-1	1	11.8	29	5.55	17.6	0.791
3	factorial	-1	-1	1	-1	4.5	25.9	5.65	25.1	1.33
4	factorial	-1	-1	1	1	5.38	16.6	5.38	20	1.37
5	factorial	-1	1	-1	-1	10.4	40.5	4.23	25.6	0.803
6	factorial	-1	1	-1	1	10.1	30.7	4.17	19.8	0.707
7	factorial	-1	1	1	-1	5.12	27.1	4.63	30.3	1.24
8	factorial	-1	1	1	1	6.13	17.4	4.64	21.8	1.41
9	factorial	1	-1	-1	-1	8.92	22	6.79	22.9	1.5
10	factorial	1	-1	-1	1	5.89	18.7	8.81	13.6	1.53
11	factorial	1	-1	1	-1	8.05	18.7	6.11	19.3	1.99
12	factorial	1	-1	1	1	5.17	18.7	7.34	12.3	2.15
13	factorial	1	1	-1	-1	5.41	28.7	6.97	28.5	1.45
14	factorial	1	1	-1	1	2.84	29.1	7.36	18.5	1.7

Run	Type	A	B	E	C	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
15	factorial	1	1	1	-1	6.2	25.4	5.95	26.5	2.08
16	factorial	1	1	1	1	4.19	27.2	6.8	17.2	2.01
17	axial	-1	0	0	0	7.82	26.1	4.59	24.2	1.15
18	axial	1	0	0	0	6.07	23.7	7.72	19.3	1.82
19	axial	0	-1	0	0	8.77	24.3	6.75	19.9	1.51
20	axial	0	1	0	0	7.55	27.9	5.51	26.8	1.32
21	axial	0	0	-1	0	10.3	30.8	6.11	20.8	1.3
22	axial	0	0	1	0	6.89	22.5	5.54	22.4	2.06
23	axial	0	0	0	-1	8.48	27.1	6.39	25.7	1.5
24	axial	0	0	0	1	8.39	23.5	5.94	19.2	1.45
25	center	0	0	0	0	7.88	23	5.75	20.5	1.52
26	center	0	0	0	0	7.95	27.3	6.32	22	1.6
27	center	0	0	0	0	7.35	25	5.6	22.4	1.55
28	center	0	0	0	0	7.27	28.2	4.96	21.1	1.37

Appendix C. Full effect and coefficient tables

Table 13.4 to Table 13.8 give every screening effect estimate for the five responses, sorted by absolute effect. Table 13.9 to Table 13.11 give the response-surface coefficient tables for the three responses not shown in §5.3.

Table 13.4: All screening effects on afucosylation.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AE	2.46	1.23	0.266	4.62	0.0191	*
E	-2.37	-1.19	0.266	-4.46	0.0209	*
AC	-1.76	-0.88	0.266	-3.31	0.0455	*
A	-1.57	-0.785	0.266	-2.95	0.06	
B	-1.34	-0.671	0.266	-2.53	0.0858	
D	-1.33	-0.664	0.266	-2.5	0.0878	
AB	-1.28	-0.64	0.266	-2.41	0.0952	
C	-0.73	-0.365	0.266	-1.37	0.264	
BD	0.452	0.226	0.266	0.85	0.458	
BE	0.362	0.181	0.266	0.681	0.545	
AD	-0.175	-0.0876	0.266	-0.329	0.764	
BC	-0.171	-0.0856	0.266	-0.322	0.769	
CD	-0.147	-0.0736	0.266	-0.277	0.8	
DE	0.0358	0.0179	0.266	0.0674	0.951	
CE	0.0118	0.00591	0.266	0.0222	0.984	

Table 13.5: All screening effects on galactosylation.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
E	-7.82	-3.91	0.193	-20.2	0.000264	***
AE	6.9	3.45	0.193	17.9	0.000382	***
C	-5.79	-2.9	0.193	-15	0.000644	***
A	-4.56	-2.28	0.193	-11.8	0.00131	**
BD	3.7	1.85	0.193	9.57	0.00242	**
AC	3.6	1.8	0.193	9.32	0.00261	**
B	3.54	1.77	0.193	9.17	0.00274	**
AB	3.28	1.64	0.193	8.49	0.00343	**
D	-2.45	-1.23	0.193	-6.35	0.0079	**
CD	2.18	1.09	0.193	5.64	0.011	*

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
DE	1.93	0.963	0.193	4.98	0.0155	*
BE	1.49	0.747	0.193	3.87	0.0306	*
AD	-0.195	-0.0974	0.193	-0.504	0.649	
BC	-0.131	-0.0657	0.193	-0.34	0.756	
CE	0.119	0.0596	0.193	0.309	0.778	

Table 13.6: All screening effects on high mannose.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.51	0.756	0.155	4.89	0.0164	*
B	-1.39	-0.696	0.155	-4.5	0.0205	*
C	-0.446	-0.223	0.155	-1.44	0.245	
BC	0.369	0.185	0.155	1.19	0.318	
E	0.158	0.0791	0.155	0.511	0.645	
D	-0.139	-0.0693	0.155	-0.448	0.685	
AC	0.0897	0.0449	0.155	0.29	0.791	
AE	0.0894	0.0447	0.155	0.289	0.792	
DE	-0.0718	-0.0359	0.155	-0.232	0.831	
CE	-0.0552	-0.0276	0.155	-0.179	0.87	
BE	0.046	0.023	0.155	0.149	0.891	
AB	0.0432	0.0216	0.155	0.14	0.898	
AD	-0.0239	-0.012	0.155	-0.0774	0.943	
CD	-0.0232	-0.0116	0.155	-0.075	0.945	
BD	-0.000576	-0.000288	0.155	-0.00186	0.999	

Table 13.7: All screening effects on acidic charge variants.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	-8.15	-4.08	0.239	-17	0.000441	***
A	-4.17	-2.08	0.239	-8.7	0.00319	**
B	2.97	1.48	0.239	6.19	0.00848	**
D	-2.55	-1.28	0.239	-5.34	0.0129	*
AB	1.98	0.992	0.239	4.15	0.0255	*
AC	-1.93	-0.964	0.239	-4.03	0.0275	*
BE	1.73	0.865	0.239	3.61	0.0364	*
AD	-0.669	-0.335	0.239	-1.4	0.256	
CE	-0.494	-0.247	0.239	-1.03	0.378	
CD	-0.492	-0.246	0.239	-1.03	0.379	
BD	-0.215	-0.108	0.239	-0.449	0.684	
BC	0.151	0.0757	0.239	0.316	0.773	
DE	0.0324	0.0162	0.239	0.0676	0.95	
E	0.0235	0.0117	0.239	0.049	0.964	
AE	0.0226	0.0113	0.239	0.0472	0.965	

Term	Effect	Coef.	Std. err.	t	p-value	Sig.

Table 13.8: All screening effects on aggregate.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.834	0.417	0.0207	20.1	0.000268	***
E	0.578	0.289	0.0207	13.9	0.0008	***
BE	-0.141	-0.0705	0.0207	-3.4	0.0424	*
C	0.116	0.058	0.0207	2.8	0.0678	
CD	-0.0742	-0.0371	0.0207	-1.79	0.171	
AD	-0.0666	-0.0333	0.0207	-1.61	0.206	
AB	-0.0593	-0.0297	0.0207	-1.43	0.248	
DE	-0.0554	-0.0277	0.0207	-1.34	0.274	
D	-0.0548	-0.0274	0.0207	-1.32	0.278	
BD	-0.0547	-0.0274	0.0207	-1.32	0.278	
AE	0.0416	0.0208	0.0207	1	0.39	
BC	-0.0235	-0.0118	0.0207	-0.567	0.61	
CE	0.0222	0.0111	0.0207	0.537	0.629	
B	-0.00671	-0.00336	0.0207	-0.162	0.882	
AC	-0.00663	-0.00332	0.0207	-0.16	0.883	

Table 13.9: Response-surface coefficients for high mannose.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	5.89	0.163	36	2.06e-14	***
A	1.09	0.111	9.77	2.36e-07	***
B	-0.421	0.111	-3.78	0.00227	**
E	-0.188	0.111	-1.69	0.115	
C	0.212	0.111	1.91	0.0788	
AB	0.148	0.118	1.26	0.231	
AE	-0.29	0.118	-2.46	0.0289	*
AC	0.293	0.118	2.48	0.0276	*
BE	0.0874	0.118	0.74	0.472	
BC	-0.118	0.118	-0.996	0.337	
EC	-0.0402	0.118	-0.341	0.739	
A^2	0.116	0.294	0.393	0.701	
B^2	0.094	0.294	0.32	0.754	
E^2	-0.215	0.294	-0.731	0.478	
C^2	0.126	0.294	0.43	0.674	

Table 13.10: Response-surface coefficients for acidic charge variants.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	22	0.542	40.6	4.43e-15	***
A	-2.01	0.369	-5.44	0.000113	***
B	1.9	0.369	5.14	0.000191	***
E	-0.124	0.369	-0.335	0.743	
C	-4.09	0.369	-11.1	5.4e-08	***
AB	1.11	0.392	2.84	0.0139	*
AE	-0.793	0.392	-2.03	0.0638	
AC	-0.248	0.392	-0.633	0.538	
BE	0.655	0.392	1.67	0.118	
BC	-0.0111	0.392	-0.0283	0.978	
EC	0.465	0.392	1.19	0.256	
A^2	-0.586	0.975	-0.601	0.558	
B^2	1.03	0.975	1.06	0.311	
E^2	-0.718	0.975	-0.736	0.475	
C^2	0.13	0.975	0.133	0.896	

Table 13.11: Response-surface coefficients for aggregate.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	1.52	0.0341	44.8	1.25e-15	***
A	0.367	0.0232	15.8	7.26e-10	***
B	-0.0157	0.0232	-0.675	0.512	
E	0.281	0.0232	12.1	1.91e-08	***
C	0.0225	0.0232	0.971	0.349	
AB	0.0135	0.0246	0.547	0.593	
AE	-0.0106	0.0246	-0.43	0.674	
AC	0.0179	0.0246	0.727	0.48	
BE	-0.00713	0.0246	-0.29	0.777	
BC	0.00414	0.0246	0.168	0.869	
EC	0.0105	0.0246	0.426	0.677	
A^2	-0.0533	0.0613	-0.87	0.4	
B^2	-0.119	0.0613	-1.93	0.0752	
E^2	0.143	0.0613	2.33	0.0366	*
C^2	-0.0626	0.0613	-1.02	0.326	

Appendix D. Analytical methods summary

Table 13.12 maps each quality attribute set at this step to the validated method that measures it and to the unit in which the result is reported. The controlled documents are listed in Table 3.1.

Table 13.12: Quality attributes, validated methods and reportable units.

Quality attribute	Validated method	Method reference	Reportable unit
Afucosylation	N-Glycan Map (2-AB HILIC-UPLC)	AMV-3010	% of glycans
Galactosylation (total %Gal)	N-Glycan Map (2-AB HILIC-UPLC)	AMV-3010	% (G1+G2 equiv.)
High Mannose	N-Glycan Map (2-AB HILIC-UPLC)	AMV-3010	% of glycans
Aggregates (HMW)	Size-Variants (SEC-HPLC)	AMV-3011	% HMW (SEC)
Acidic charge variants (deamidation)	Charge Variants (icIEF)	AMV-3013	% (CEX/icIEF)
Host Cell Protein (HCP)	Host-Cell Protein ELISA	AMV-3012	ng/mg
Residual DNA	Residual DNA (qPCR)	AMV-3014	ng/dose

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